

An even log-concave counterexample to the Gaussian Grünbaum bound

Autonomous mathematical discovery packet
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Status. Candidate full counterexample, likely valid, pending human review. The proof is exact and has no computational dependency.

1 Source question

Fradelizi, Langharst, Liu, Marín Sola, and Tang [1] prove a one-dimensional Gaussian-form Grünbaum inequality for γ -transport-concave measures. If $B = (a, b)$ has mass $t > 0$ and conditional barycenter

$$g_B = \frac{\int_a^b x d\mu(x)}{\mu(B)},$$

their Corollary 5.2 gives

$$\mu((a, g_B]) \geq \Phi\left(-\frac{I_\gamma(t)}{t}\right), \quad I_\gamma(t) = \varphi(\Phi^{-1}(t)), \quad (1)$$

where Φ and φ are the standard Gaussian distribution function and density. On page 30 they ask whether the same conclusion might hold for measures that are even. Figure 1 records the exact passage.

$$\Phi^{-1}(\mu((a, g))) \geq (\Phi^{-1} \circ \Phi_\mu) \left(\int_0^t \Phi_\mu^{-1}(r) \frac{dr}{t} \right) \geq \int_0^t \Phi^{-1}(r) \frac{dr}{t}.$$

Recall that a measure μ is convex if and only if the reciprocal of its density is convex. Letting ψ be defined by (76), one can take two derivatives of its reciprocal and see that γ -transport concave measures are not necessarily convex measures.

We finish by showing that a measure μ that is even and γ -transport concave must be centered Gaussian with arbitrary variance. Although this does not necessarily discredit the possibility of Corollary 5.2 holding for measures that are even, as being γ -transport concave is sufficient but not necessary; we do not know how to relate the two barycenters without this condition.

Proposition 5.4. *Let μ be an even probability measure on $\mathbb{R}$ with density. Then, it is γ -transport concave if and only if there exists $\sigma > 0$ such that μ has density*

$$\frac{d\mu(r)}{dr} = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{r^2}{2\sigma^2}}.$$

Proof. Let T be the optimal transport map from γ onto μ , i.e. T satisfies $\Phi_\mu(T(r)) = \Phi(r)$ and goes from $\mathbb{R}$ to $\text{supp}(\mu)$. First, we have that T must be odd: we see that

Figure 1: The open-problem signal in arXiv:2507.06759v2, page 30.

2 Proof intuition

On the negative half-line the centered Laplace density is exponential. Its conditional mean lies one scale length from the origin, leaving exactly an e^{-1} fraction of the half-line mass farther into the tail. This is the extremal one-dimensional log-concave Grünbaum behavior, and it is smaller than the stronger Gaussian expression in (1). A sufficiently long but finite interval retains the strict gap. Taking $(-5, 0)$ makes the comparison elementary: the two sides can be separated at the rational number $1/5$.

3 Counterexample

Theorem 1. *Let μ be the centered Laplace probability measure on $\mathbb{R}$,*

$$d\mu(x) = \frac{1}{2}e^{-|x|} dx,$$

and let $B = (-5, 0)$. Then μ is even, log-concave, and a convex measure in the sense of [1]. If $t = \mu(B)$ and $g = g_B$, then

$$\mu((-5, g]) < \frac{1}{5} < \Phi\left(-\frac{I_\gamma(t)}{t}\right).$$

Consequently, the proposed extension of (1) to even measures is false, even within the class of even log-concave convex measures.

Proof. The density is even, and

$$\log\left(\frac{1}{2}e^{-|x|}\right) = -\log 2 - |x|$$

is concave. Its reciprocal $2e^{|x|}$ is convex, so μ is also a convex measure under the source definition.

On $(-5, 0)$ the density is $e^x/2$. Direct integration gives

$$t = \frac{1 - e^{-5}}{2}, \tag{2}$$

$$g = \frac{\frac{1}{2} \int_{-5}^0 x e^x dx}{t} = -1 + \frac{5}{e^5 - 1}, \tag{3}$$

$$\mu((-5, g]) = \frac{e^g - e^{-5}}{2}. \tag{4}$$

In particular $-1 < g < 0$, so the conditional barycenter lies in B .

We first bound (4). From $e > 1 + 1 + \frac{1}{2} + \frac{1}{6} = \frac{8}{3}$ we obtain $e^5 > 101$. Hence

$$g = -1 + \frac{5}{e^5 - 1} < -1 + \frac{5}{100} = -\frac{19}{20}.$$

The exponential series also gives

$$e^{19/20} > 1 + \frac{19}{20} + \frac{1}{2} \left(\frac{19}{20}\right)^2 + \frac{1}{6} \left(\frac{19}{20}\right)^3 = \frac{122119}{48000} > \frac{5}{2}.$$

Therefore $e^g < e^{-19/20} < 2/5$, and (4) yields

$$\mu((-5, g]) < \frac{e^g}{2} < \frac{1}{5}. \tag{5}$$

It remains to bound the Gaussian side. Put $x = I_\gamma(t)/t$. Since $I_\gamma(t) \leq \varphi(0) = 1/\sqrt{2\pi} < 2/5$ and, by $e^5 > 101$, equation (2) gives $t > 50/101$, we have

$$x < \frac{2/5}{50/101} = \frac{101}{125} =: c. \tag{6}$$

Here we used the elementary Archimedean estimate $\pi > 25/8$.

For $y \geq 0$ one has $e^{-y} \leq 1 - y + y^2/2$ (differentiate the difference twice). Thus

$$e^{-u^2/2} \leq 1 - \frac{u^2}{2} + \frac{u^4}{8}.$$

Moreover,

$$c - \frac{c^3}{6} + \frac{c^5}{40} = \frac{2668545614003}{3662109375000} < \frac{3}{4}.$$

Using monotonicity of Φ and (6),

$$\begin{aligned} \Phi(-x) &> \Phi(-c) \\ &= \frac{1}{2} - \frac{1}{\sqrt{2\pi}} \int_0^c e^{-u^2/2} du \\ &> \frac{1}{2} - \frac{2}{5} \left(c - \frac{c^3}{6} + \frac{c^5}{40} \right) > \frac{1}{2} - \frac{2}{5} \cdot \frac{3}{4} = \frac{1}{5}. \end{aligned}$$

Together with (5), this proves the strict reverse of (1). □

4 Smooth positive counterexamples

The example is already positive and continuous, but its density is not differentiable at the origin. This is inessential. Let $\mu_\varepsilon = \mu * \gamma_\varepsilon$, where γ_ε is a centered Gaussian of variance ε^2 . The density of μ_ε is positive, smooth, even, and log-concave; log-concavity follows from the Prékopa–Leindler theorem [2]. A positive log-concave density has a convex reciprocal, so these are convex measures as well.

As $\varepsilon \downarrow 0$, the densities converge to the Laplace density in L^1 . On the fixed bounded interval B , both its mass and its truncated first moment converge. Hence $g_{B,\varepsilon} \rightarrow g_B$, and the mass of $(-5, g_{B,\varepsilon}]$ converges to the left side of Theorem 1. The Gaussian expression is continuous in $t \in (0, 1)$. The strict gap in the theorem therefore persists for every sufficiently small $\varepsilon > 0$.

5 Verification, scope, and novelty

The accompanying script evaluates the closed formulas at 80 decimal digits and checks the rational polynomial bound; it is a regression test, not part of the proof. A bounded search through 11 August 2026 covered the run indexes, exact arXiv id and title, the exact source phrase, current arXiv v2, publication metadata, and combinations of *even measure*, *transport concavity*, *Laplace*, and *Grünbaum*. No later answer or matching counterexample was found.

This packet refutes the proposed extension under evenness, including the stronger even log-concave/convex-measure variant. It does not weaken the source’s proved theorem under γ -transport concavity; indeed, the source proves that an even γ -transport-concave density must be Gaussian, while the Laplace density is not Gaussian.

References

- [1] M. Fradelizi, D. Langharst, J. Liu, F. Marín Sola, and S. Tang, *Grünbaum’s inequality for Gaussian and convex probability measures*, arXiv:2507.06759v2 (2025), especially Corollary 5.2 and p. 30.
- [2] A. Prékopa, *Logarithmic concave measures with applications to stochastic programming*, Acta Sci. Math. (Szeged) 32 (1971), 301–316.